

Recap

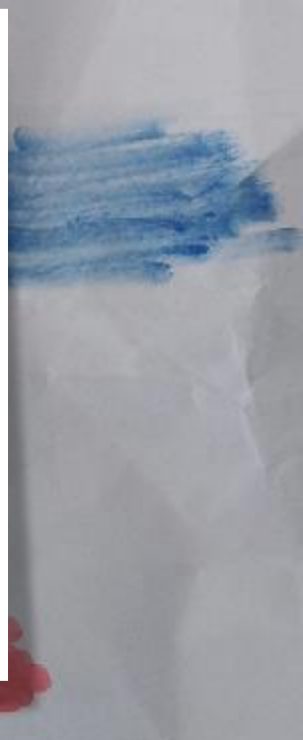
- Workflow of drug discovery
- Target, Hit, Lead, Candidate
- Clinical trials

CADD - 2026

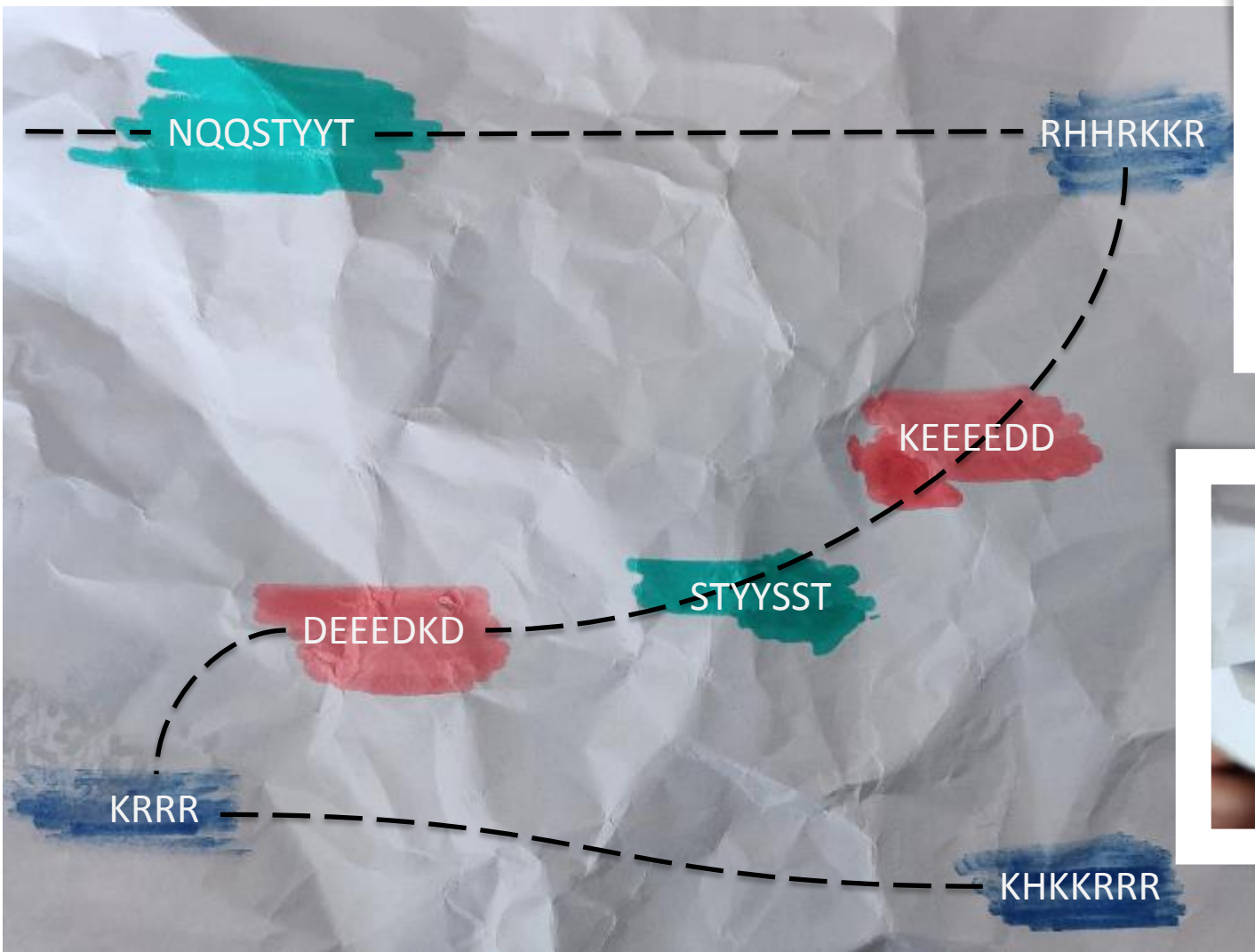
Target: Protein

Outline

- Basic concepts of protein 3D structure
- Concepts dictating protein-ligand interactions
- Introduction to the static and dynamic model of a protein



Protein



 Polar

 Positively charged

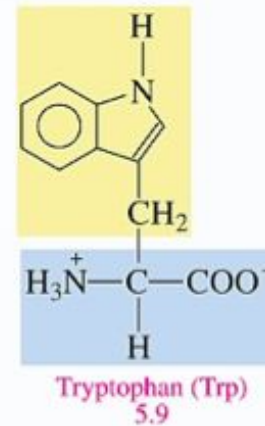
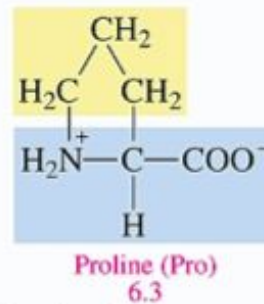
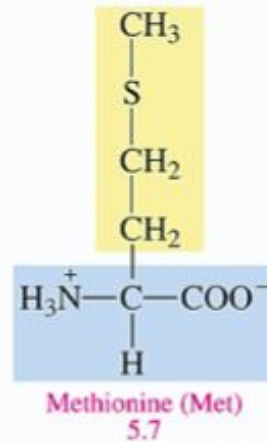
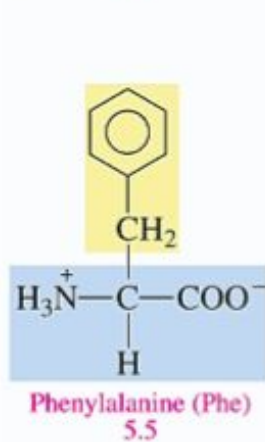
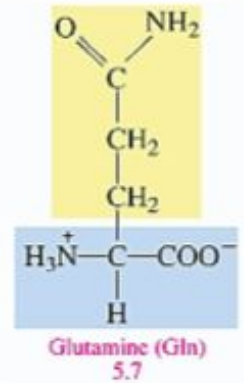
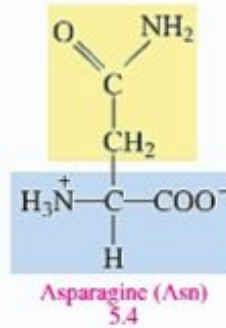
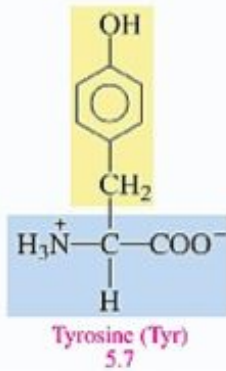
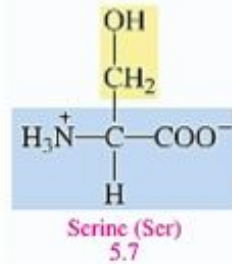
 Negatively charged

Protein – one dimension

POLAR AMINO ACIDS				NONPOLAR AMINO ACIDS			
AMINO ACID			SIDE CHAIN	AMINO ACID			SIDE CHAIN
Aspartic acid	Asp	D	negative	Alanine	Ala	A	nonpolar
Glutamic acid	Glu	E	negative	Glycine	Gly	G	nonpolar
Arginine	Arg	R	positive	Valine	Val	V	nonpolar
Lysine	Lys	K	positive	Leucine	Leu	L	nonpolar
Histidine	His	H	positive	Isoleucine	Ile	I	nonpolar
Asparagine	Asn	N	uncharged polar	Proline	Pro	P	nonpolar
Glutamine	Gln	Q	uncharged polar	Phenylalanine	Phe	F	nonpolar
Serine	Ser	S	uncharged polar	Methionine	Met	M	nonpolar
Threonine	Thr	T	uncharged polar	Tryptophan	Trp	W	nonpolar
Tyrosine	Tyr	Y	uncharged polar	Cysteine	Cys	C	nonpolar

Amino acids

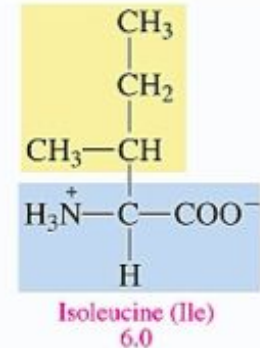
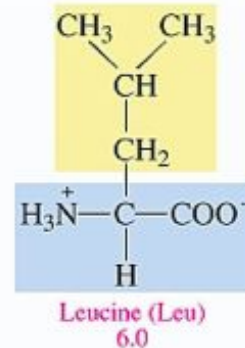
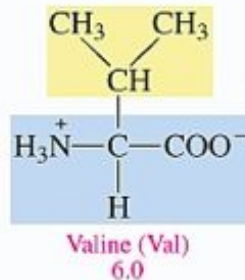
The polar aminoacids



less polar

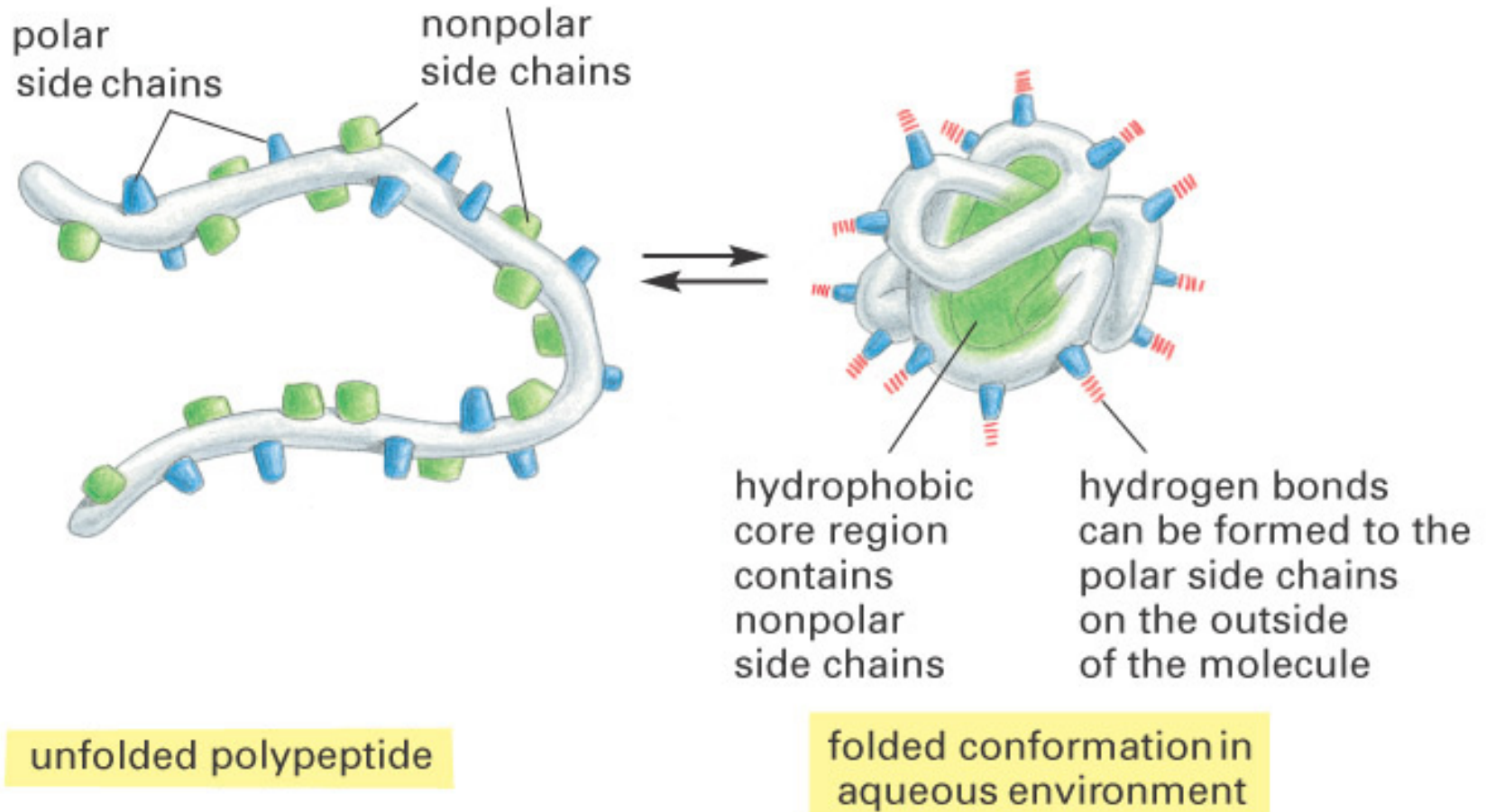
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non-polar



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Protein – one dimension to three dimension



Hierarchical Architecture in Proteins

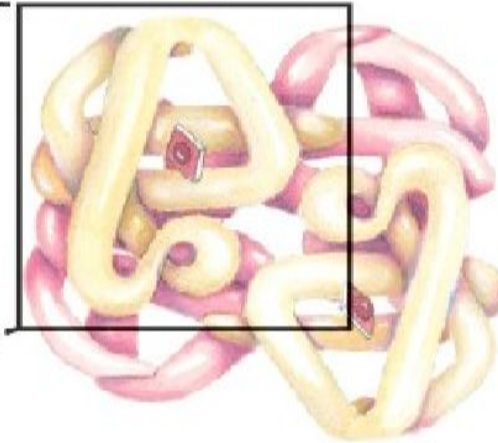
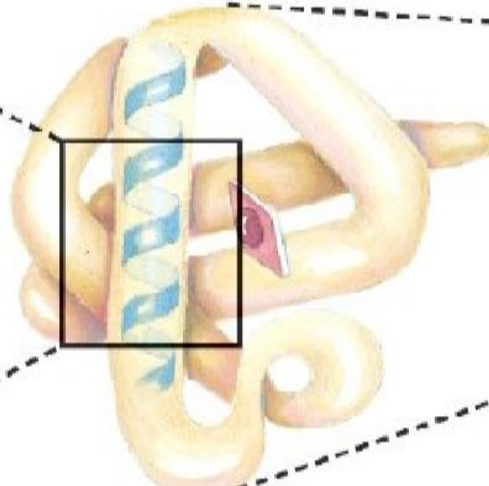
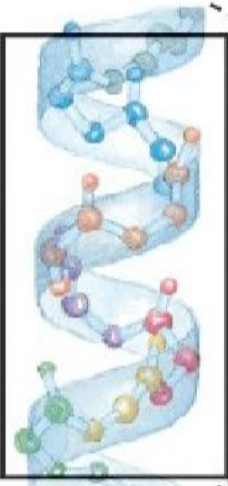
Primary
Structure

Secondary
Structure

Tertiary
Structure

Quaternary
Structure

Lys
Lys
Gly
Gly
Leu
Val
Ala
His



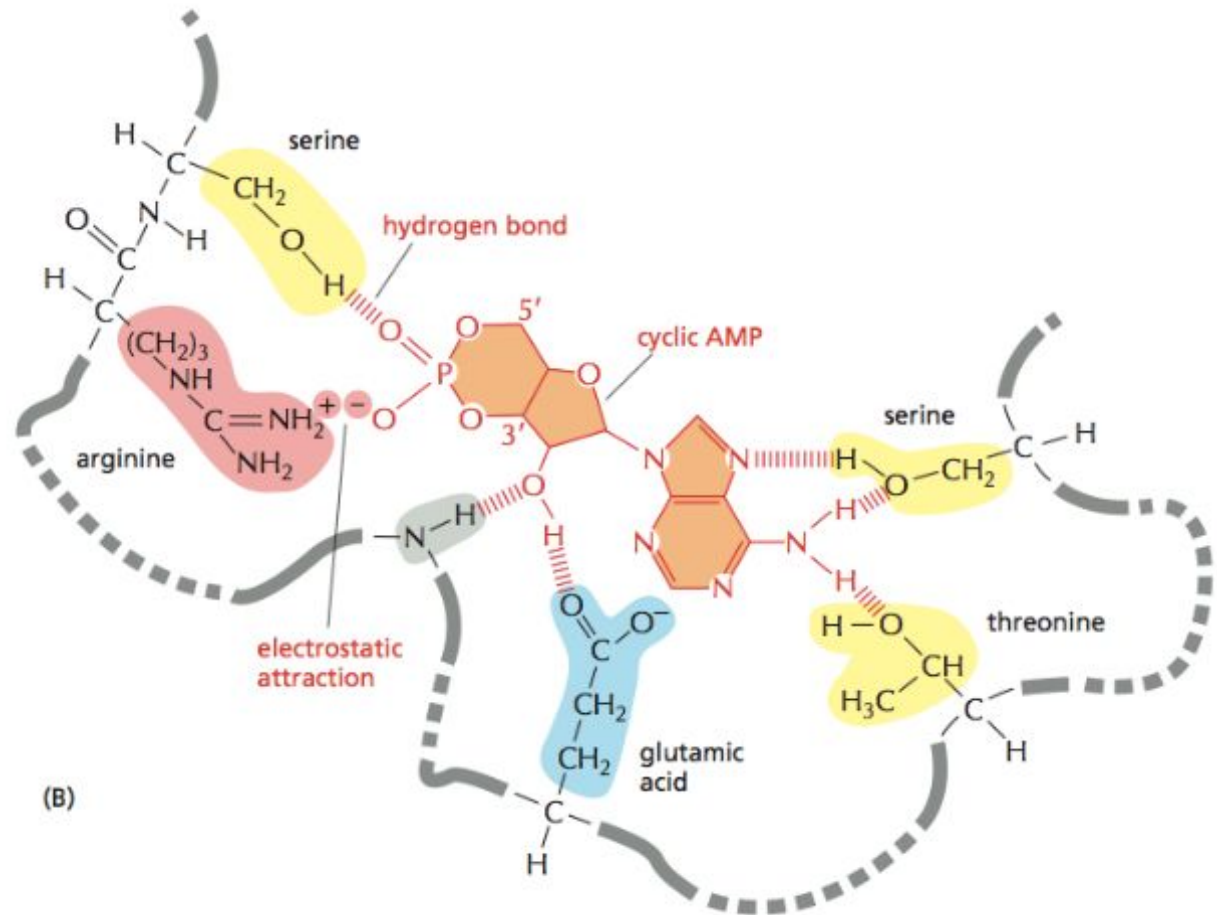
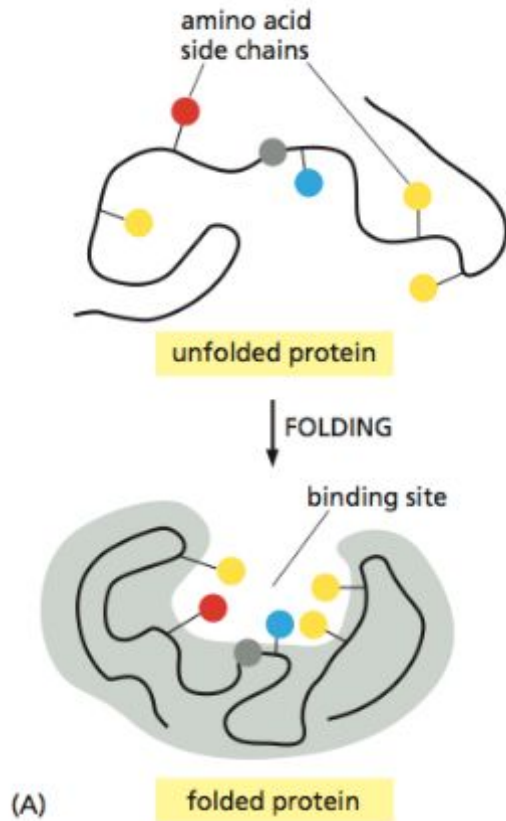
Residues

Helix

Chain

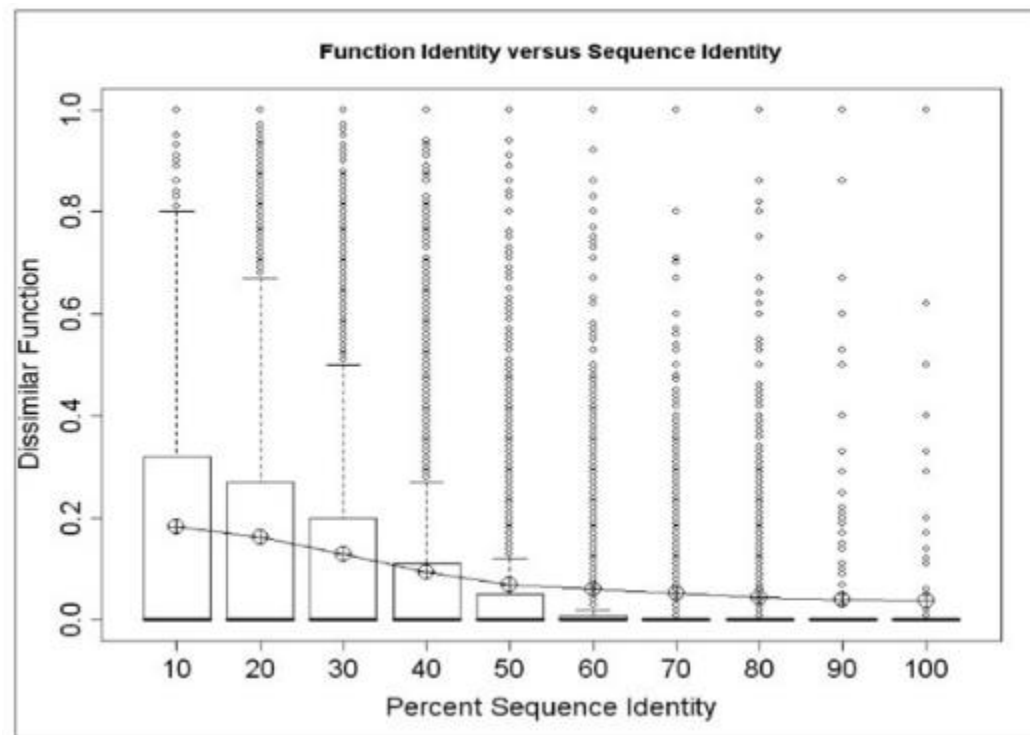
Subunits

The Surface Conformation of a Protein Determines Its Chemistry



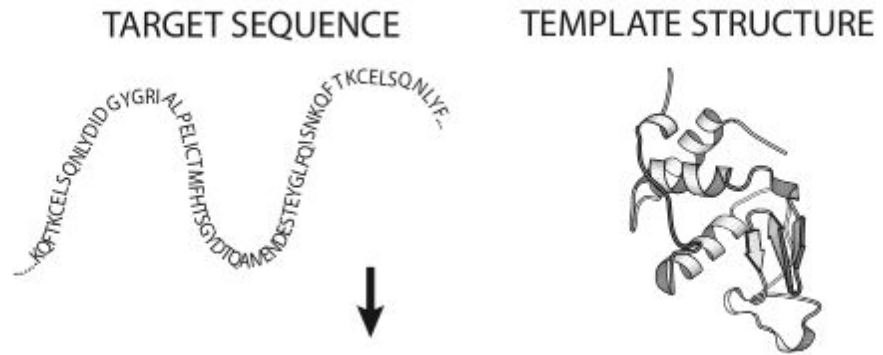
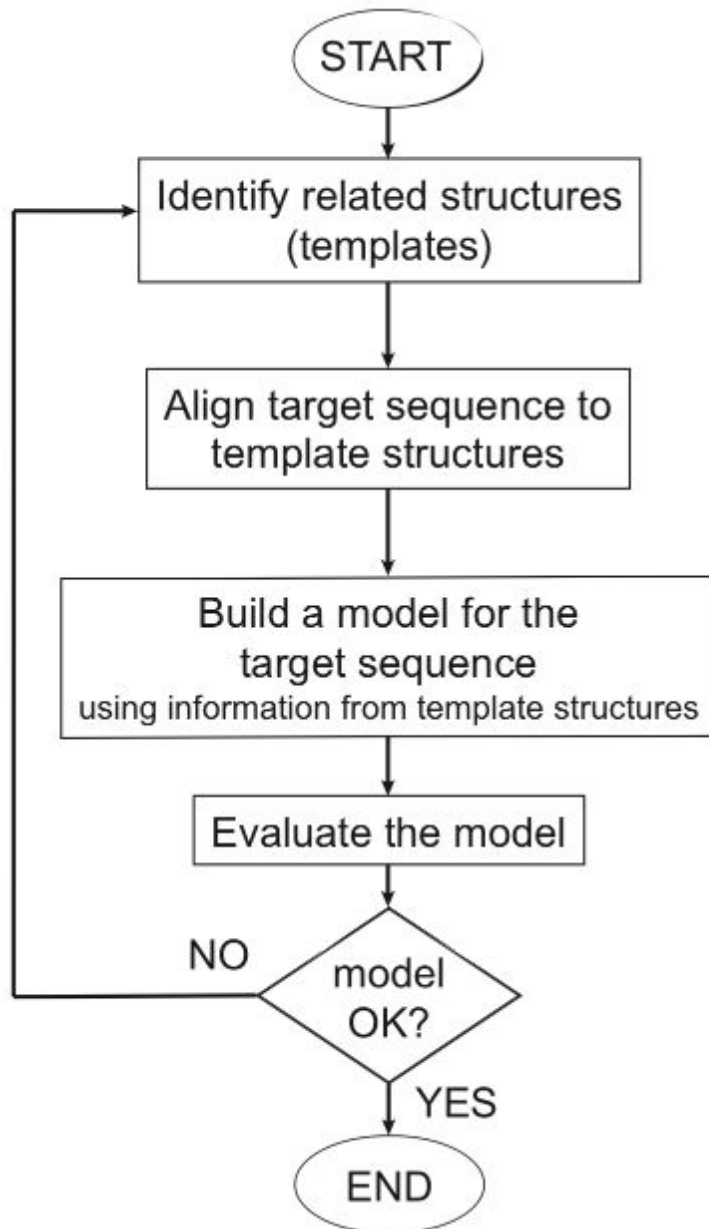
Protein structure availability

- Experimentally resolved structures: Crystal, Cryo-EM, NMR (<https://www.rcsb.org/>)
- In case of unavailability...



Sangar, V., Blankenberg, D.J., Altman, N. and Lesk, A.M., 2007. Quantitative sequence-function relationships in proteins based on gene ontology. *BMC bioinformatics*, 8(1), p.294.

Homology modeling: usual approach



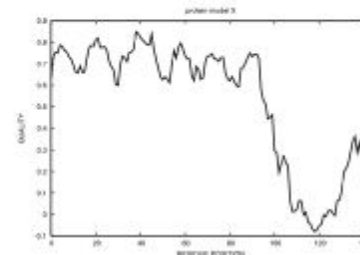
ALIGNMENT

TARGET ...SCDKLLDDELDDDIACAKKILAIKGID...
TEMPLATE ...SCDKFLDDITDDIMCAKKILDIKGID...

TARGET
MODEL



QUALITY
PROFILE



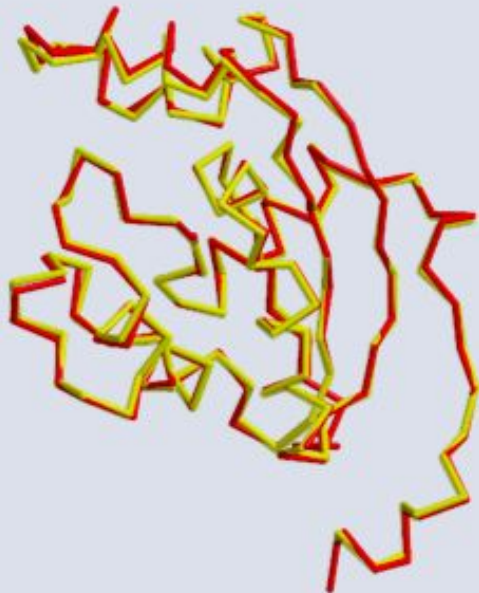
More to be taught
in subsequent weeks..

Model Accuracy

High accuracy

NM23
Seq id 77%

C α equiv 147/148
RMSD 0.41Å



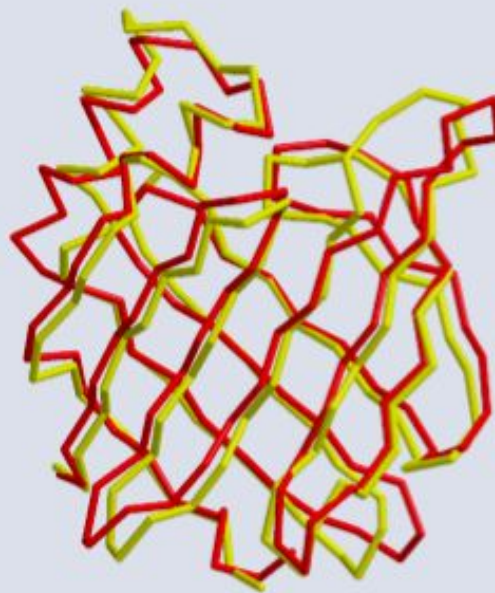
Side chains
Core backbone
Loops

X-RAY / MODEL

Medium accuracy

CRABP
Seq id 41%

C α equiv 122/137
RMSD 1.34Å



Side chains
Core backbone
Loops
Alignment

Low accuracy

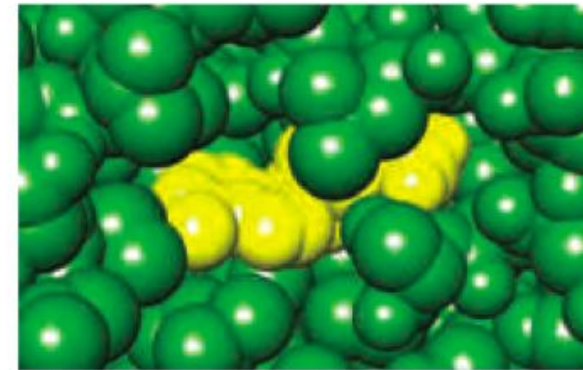
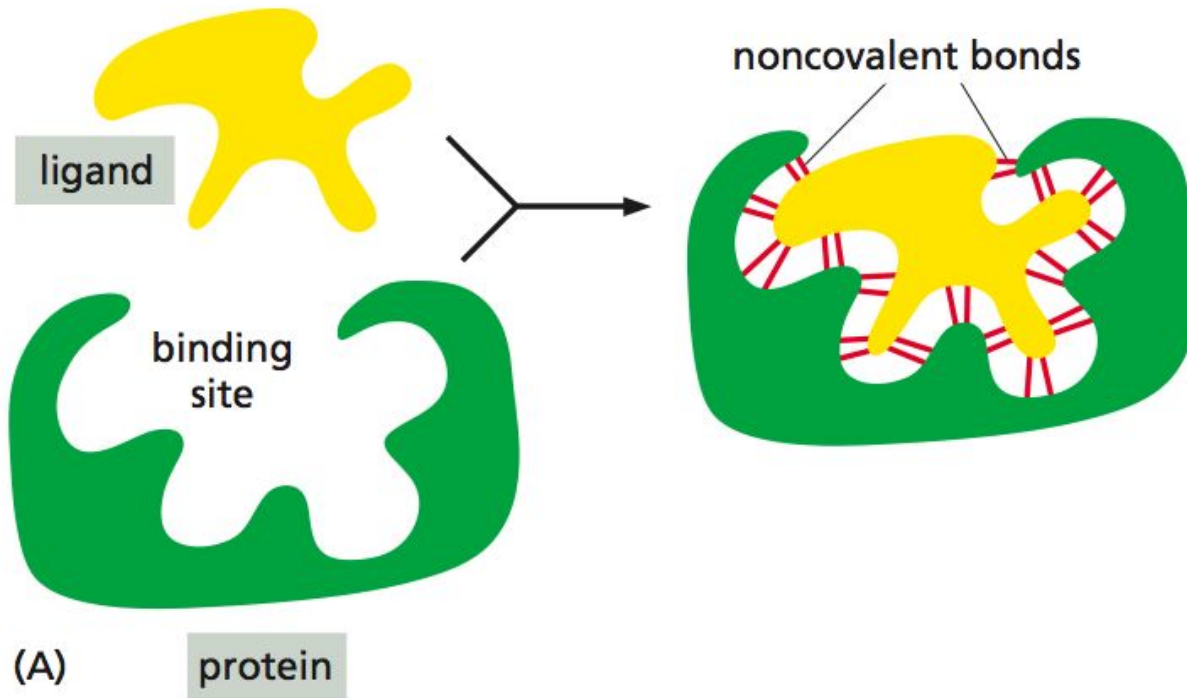
EDN
Seq id 33%

C α equiv 90/134
RMSD 1.17Å



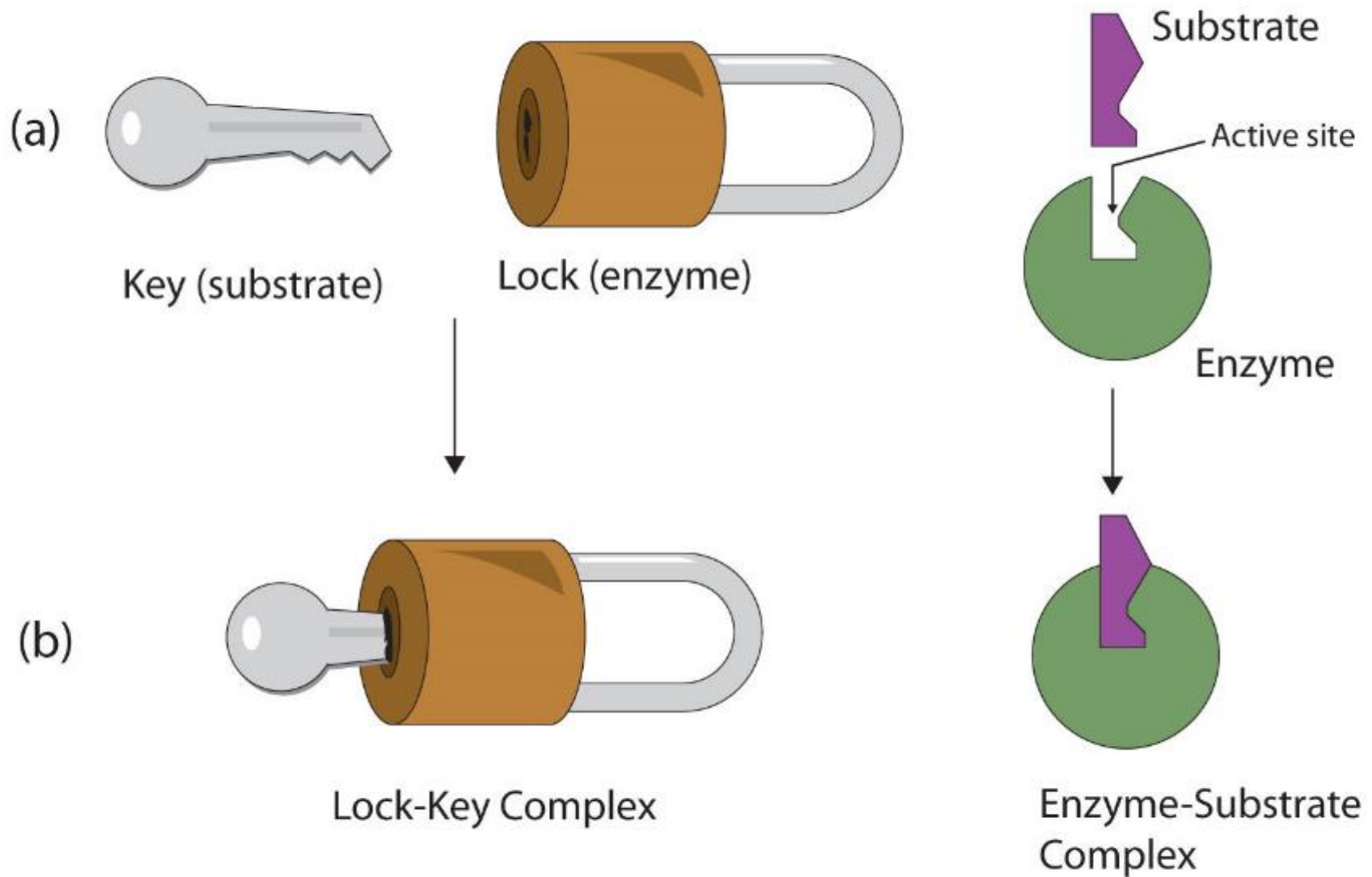
Side chains
Core backbone
Loops
Alignment
Fold assignment

Complementarity in protein-ligand interaction

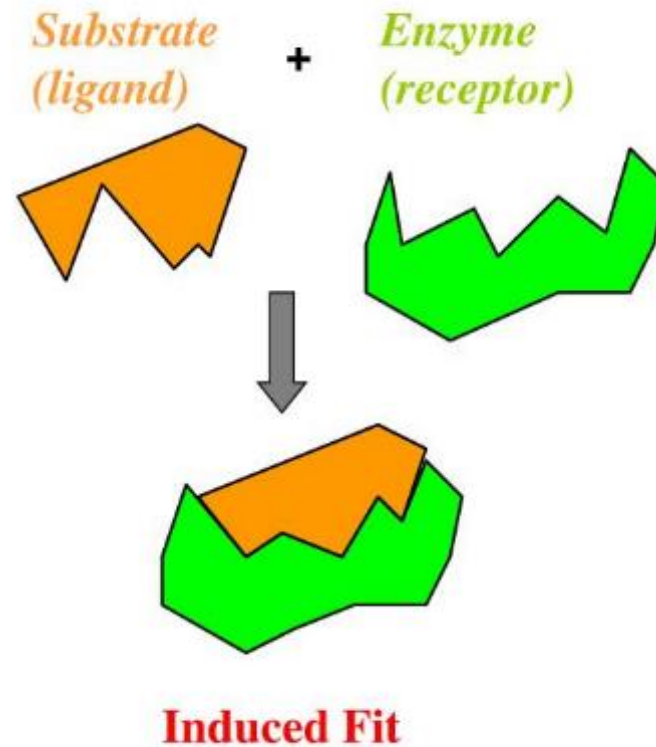
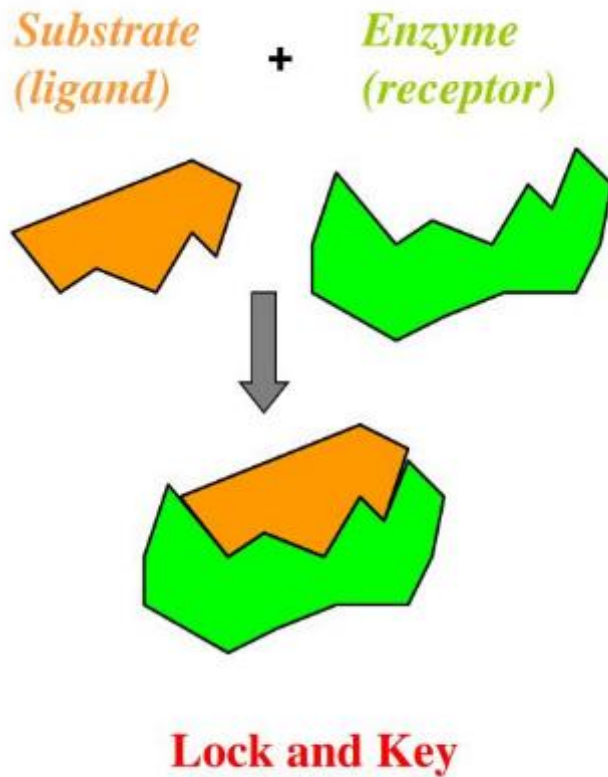


Ligand: protein/substrate/molecule

Complementarity in protein-ligand interaction (Shape)



Complementarity in substrate-enzyme interaction



Docking

Rigid Docking

- This approximation treats both the ligand and the receptor as rigid and

Flexible Docking

- A more common approach is to model the ligand flexibility while

In biology, interactions are dynamic

ligand binding), are dealt with by rigid protein docking.

- Computationally inexpensive

induced fit docking is used with flexible binding sites

- Computationally expensive

Atoms never stop jiggling

- In real life, atoms are in constant motion.
 - They will *not* go to an energy minimum and stay there.
- Given enough time, the simulation samples the Boltzmann distribution
 - That is, the probability of observing a particular arrangement of atoms is a function of the potential energy
 - In reality, one often does not simulate long enough to reach all energetically favorable arrangements

